

Journal Supplementary E — Full dataset descriptors

Companion to: *Beyond Accuracy* (Section V)

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A.1 Scope

This supplement expands Section V with the complete provenance and per-subset technical inventory of the eleven evaluation entities: six labelled HSI scenes (treated in conference supplement C and summarised below for cross-reference) plus five HIDSAG mineralogical subsets (treated here at full depth).

B.2 Labelled HSI scenes

Six standard labelled benchmarks from the UPV/EHU Hyperspectral Remote Sensing Scenes collection: Indian Pines (AVIRIS, 200 bands, 2812 documents in the canonical fit, 12 topics), Salinas (AVIRIS, 204, 3520, 12), Salinas-A (subset of Salinas, 204, 1320, 6), Pavia University (ROSIS-03, 103, 1980, 9; VNIR-only sensor), Kennedy Space Center (AVIRIS-Old, 176, 2686, 12), Botswana (EO-1 Hyperion, 145, 2775, 12). See conference supplement C for per-scene sensor and class-catalogue details.

C.3 HIDSAG mineralogical subsets

Source. HIDSAG is a curated industrial database of hyperspectral point-spectrum measurements over mineralogical samples acquired with a SWIR sensor (the `swir_low` modality consumed in this paper covers $\sim 1100\text{--}2500$ nm in 268 contiguous narrow bands). The database exposes five thematic subsets distributed as zip archives.

C.3.1 GEOCHEM

Content. 28 samples, 106 crop ROIs, 318 hyperspectral cubes, 18 geochemical target variables: Al, As, Ca, Cu, Fe, K, Mo, Nb, P, Pb, Rb, S, Si, Sr, Ti, Y, Zn, Zr (in ppm or wt-%, depending on element). Each sample carries a per-ROI cube and per-ROI geochemistry vector; the document construction averages cube pixels within each ROI to produce one document per (sample, crop, measurement).

C.3.2 GEOMET

Content. 146 samples (the largest by sample count), 146 crops, 438 cubes, 5 geometallurgical target variables: Cu rec (% copper recovery), Lime cons (kg/t lime consumption), Mo rec (% molybdenum recovery), PH (pulp pH), WI (Bond work index in kWh/t). The targets are continuous regression labels rather than classification labels; the band-mask diagnostic on GEOMET uses between-covariate-variance band ranking against these continuous targets.

C.3.3 MINERAL1

Content. 99 samples, 99 crops, 297 cubes, 33 mineral abundance target variables: Actinolite, Albite, Andesina (An40), Anhydrite/Gypsum, Anorthite, Apatite, Biotite, Bornite, Bytownite (An80), Calcite, Chalcopyrite, Chalcosite/Digenite, Chlorite, Covelite, Epidote, Fe Oxides, Fe-Clays, K-Feldspar, Kaolinite, Labradorite (An60), Molybdenite, Muscovite/Sericite, Oligoclasa (An20), Other Amphibole, Others, Others Carbonates, Others Ti Minerals, Pyrite, Pyroxene, Quartz, Rutile, Siderite, Tourmaline. The list mixes feldspars (An20/40/60/80), sulfides (Bornite/Chalcopyrite/Chalcosite/Covelite), carbonates (Calcite/Siderite), and hydrothermal indicators (Tourmaline/Muscovite/Kaolinite) at $\sim$ % mineralogy per ROI.

C.3.4 MINERAL2

Content. 20 samples (smallest), 20 crops, 60 cubes, 25 mineral abundance target variables: Alunite, Ankerite, Apatite, Bornite, Calcite, Chalcocite, Chalcopyrite, Clinocllore, Enargite, Gypsum, Hematite, Illite, Kaolin group, Molybdenite, Muscovite, Orthoclase, Phengite, Pirit, Plagioclase, Pyrophyllite, Quartz, Siderite, Sphalerite, Tenantite, Tetrahedrite. MINERAL2 covers a slightly different mineralogical regime from MINERAL1, emphasising high-sulfidation epithermal mineralogy (Alunite, Pyrophyllite, Enargite). The very high topic-confidence MINERAL2 reports under the band-mask diagnostic (> 0.94 across all four masks) is consistent with the sharper mineralogical signatures in this subset.

C.3.5 PORPHYRY

Content. 28 samples, 56 crops, 168 cubes, 11 target variables: Bt (% biotite), Cal (% calcite), Esp (% specularite), Hem (% hematite), Kao (% kaolinite), Mb (% molybdenite), Mgt (% magnetite), Py (% pyrite), Qz (% quartz), [Cpy + Py + Bo] (% combined sulfides), group (categorical lithology). PORPHYRY is the most operational-decision-relevant of the five subsets: porphyry-copper deposits are the primary global Cu source and the target variables track the alteration and sulfide-distribution characteristics that drive resource estimation.

D.4 Document construction on HIDSAG

For each subset the corpus consists of one spectrum per (sample, crop, cube) tuple, computed as the per-band mean over labelled pixels within the crop ROI. The tokenisation follows the same min-max + quantisation pipeline used on the labelled scenes ($Q+1 = 13$ levels). The document count entering the canonical fit on each subset is the cube count: GEOCHEM 318, GEOMET 438, MINERAL1 297, MINERAL2 60, PORPHYRY 168.

E.5 Sensor differences from labelled scenes

HIDSAG is acquired with a point spectrometer rather than an imaging array, so the spatial-coherence assumptions that motivate Felzenszwalb / SLIC on the labelled scenes do not apply. The framework axes F-6 (cross-method spatial agreement) and F-7 (spatial topic-label coupling on the $H \times W$ raster) are therefore not run on HIDSAG; axes F-1, F-2, F-3, F-4, F-5, F-8, F-9, F-10, F-11, F-12 are all run, with F-5 using between-covariate-variance band ranking instead of the labelled Fisher discriminant.

F.6 Continuous-measurement layer

A complementary derived layer at `data/derived/hidsag_topic_measurements/<subset>.json` joins the per-document θ matrix shipped by the band-mask builder with the continuous mineralogical / geometallurgical assay values extracted from each sample's `metadata.json` (per the HIDSAG figshare

data release [1]). The layer ships 427 records across the five subsets, exposing 5–33 continuous variables per subset:

- GEOMET: Cu rec, Mo rec, PH, Lime cons, WI (5 vars, 146 records)
- MINERAL1: 33 mineral abundance % (99 records)
- MINERAL2: 25 mineral abundance % (20 records)
- GEOCHEM: 18 elemental wt%/ppm (106 records)
- PORPHYRY: 11 mineral abundance % incl. biotite / molybdenite / chalcopyrite (56 records)

This layer feeds the topic-conditional ridges, the GEOMET corner plot, and the HIDSAG mosaic of supplement I. Figure ?? of Suppl I reports the most striking signal: on PORPHYRY, per-topic mean % biotite ranges 1.1–39.6, indicating that the canonical-fit topic basis on PORPHYRY captures the porphyry-alteration mineralogy at high fidelity. The extraction builder is documented at `data-pipeline/build_hidsag_topic_measurements.py`.

G.7 Citation status

HIDSAG is cited via two complementary bibliographic entries:

- **hidsag2023** — Ehrenfeld, Egaña, Santibáñez-Leal, Garrido, Ojeda, Townley, and Navarro, “HIDSAG: Hyperspectral Image Database for Supervised Analysis in Geometallurgy”, *Scientific Data* 10:164 (2023), doi:10.1038/s41597-023-02061-x. The canonical paper introducing the database, its acquisition protocol, and the five thematic subsets.
- **hidsag2023figshare** — Ehrenfeld (2023), HIDSAG figshare Collection, doi:10.6084/m9.figshare.c.5983921.v1. The data release accompanying the paper.

The reference implementation is at <https://github.com/alges/hidsag>. The companion code repository for this manuscript consumes a curated subset documented at `data/derived/core/hidsag_curated_subset`. Felipe Santibáñez-Leal is a co-author of **hidsag2023**; the present manuscript is methodologically distinct (multi-axis evaluation framework for topic models applied to HIDSAG) and cites HIDSAG as a data source rather than re-publishing its content.

References

- [1] A. Ehrenfeld, “HIDSAG: Hyperspectral image database for supervised analysis in geometallurgy,” figshare Collection, <https://doi.org/10.6084/m9.figshare.c.5983921.v1>, 2023, data collection accompanying [?]; five thematic subsets (GEOMET, MINERAL1, MINERAL2, GEOCHEM, PORPHYRY). Reference implementation at <https://github.com/alges/hidsag>.